

Apurva Kokate

Machine Learning Researcher & Engineer | PhD Candidate

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Areas of Expertise

Machine Learning Systems • Deep Learning for Graphical, Vision & Text Data • Graph Neural Networks • Model Evaluation & Reliability • Robustness & Generalization • Scientific & Applied Machine Learning

Education

Doctor of Philosophy, Artificial Intelligence Oregon State University, Corvallis, OR, USA	2021–Present
Master of Science, Computer Science Iowa State University, Ames, IA, USA	2016–2018
Bachelor of Engineering, Computer Engineering Vidyalankar Institute of Technology, Mumbai, Maharashtra, India	2014–2018

Work Experience

Graduate Assistant <i>Location: Oregon State University</i> - TA for AI 534 (Machine Learning), CS 434 (Machine Learning & Data Mining), and CS 325 (Analysis of Algorithms) - Mentored students in algorithm design, data structures, supervised learning, model evaluation, and debugging best practices	Machine Learning Sept 2021–Present
Full Stack Software / ML Engineer <i>Location: Kingland, Iowa</i> - Co-created Kingland's enterprise Text Analytics platform, integrating deep-learning (NER), contextual language modeling, and document-parsing pipelines - Engineered cloud-native ingestion and NLP services that processed <i>5M+ financial records</i> and <i>1.1M+ daily updates</i>, enabling large-scale extraction and normalization of entities from highly variable unstructured content - Achieved 99% entity-resolution accuracy for DTCC's mutual-fund data by designing context-aware extraction workflows addressing challenges such as alias resolution, domain-specific language, and bias mitigation - Delivered measurable business value by automating manual document review, improving data quality, and enabling risk monitoring and compliance workflows described in Kingland's "Text Analytics Use Cases" series - Built end-to-end features across the full stack — microservices, APIs, data pipelines, and UI components — ensuring seamless integration of ML models into production systems serving Fortune 500 clients	Text Analytics July 2018–May 2021
Summer Software Intern <i>Location: Icon Laboratory, Iowa</i> - Designed and implemented a security module for IoT devices, incorporating obfuscation and secure key-retrieval mechanisms to improve data privacy and system reliability - Integrated lightweight cryptographic techniques tailored for resource-constrained embedded IoT environments	IoT Security May 2017 – Jul 2017

Selected Research & Publications

MOSE-GNN: A Motif-Based Self-Explaining Graph Neural Network for Molecular Property Prediction <i>Location: Oregon State University</i> - Designed MOSE-GNN, a graph-based learning architecture that improves model reliability, transparency, and generalization for molecular property prediction - Developed a molecular fragment-based interpretability framework producing <i>chemically meaningful</i> and <i>global</i> explanations - Demonstrated superior fidelity, consistency, and motif-correctness over post-hoc explainers - Enabled chemist-aligned model transparency for safety-critical molecular property prediction tasks	LoG conference 2024
Adversarial Robustness of Vision Transformers in Medical Imaging <i>Oregon State University</i>	Project 2023

- Evaluated [adversarial robustness](#) and attack transferability across CNN, hybrid CNN–Transformer, and Vision Transformer segmentation models using FGSM and DAG attacks on biomedical datasets.
- Showed that adversarial examples do not transfer across architectures and that CNN-based models are more robust, highlighting architecture-dependent failure modes critical for interpretable and trustworthy AI systems.

Reproducibility and Reliability of Deep Neural Networks

Project

Oregon State University

2022

- Analyzed training and test-time variability in deep neural networks, demonstrating that fixed random seeds and standard regularization do not guarantee reproducible representations or predictions.
- Introduced *Effective Test Accuracy*, a [reliability-aware evaluation metric](#) that captures prediction variability across runs, improving transparency and trust in model performance reporting.

Interpretable Deep Learning for Guided Microstructure–Property

Nature Partner Journals

Exploration in Photovoltaics

Location: Iowa State University

2019

- Applied interpretable deep learning to identify [microstructural attributes](#) most predictive of photovoltaic efficiency.
- Built *explainable CNN pipelines* enabling scientists to trust, validate, and utilize ML-driven insights in materials discovery.
- Integrated attribution analysis with physics-based reasoning to accelerate photovoltaic materials exploration.

A Study of Interpretability Mechanisms for Deep Convolutional Networks

ISU Digital Repository

Location: Iowa State University

2018

- Conducted a [systematic study of vision-based interpretability](#) algorithms, creating a taxonomy for pixel-level attribution
- Implemented and benchmarked saliency, perturbation, and decomposition techniques across deep CNN architectures
- Revealed strengths, limitations, and failure modes of popular interpretability approaches used in both academia and industry

A Forward–Backward Approach for Visualizing Information Flow in Deep Networks

NeurIPS

Location: Iowa State University

2017

- Co-developed a [layer-wise attribution](#) method using mathematical inversion of the forward pass to analyze information flow
- Improved interpretability and debugging of CNNs, yielding a documented *15% improvement* in classification accuracy
- Demonstrated how forward–backward signal decomposition enhances transparency and reliability in deep neural networks

Position: Explainable AI Needs Domain Knowledge Consensus to Be Useful

OpenReview

Location: Oregon State University

2025

- Authored a [position paper](#) arguing that domain knowledge consensus is essential for achieving actionable and trustworthy XAI
- Analyzed limitations of current model-agnostic explainability methods and highlighted failure cases arising from domain-agnostic interpretation
- Proposed a framework recommending *domain-aligned priors*, evaluation protocols, and ontology-driven validation for reliable XAI
- Presented guidelines advocating collaboration between ML researchers and domain experts to establish shared interpretability standards

Leadership & Service

- **Ethics Reviewer** for the Dataset and Benchmarking Track, NeurIPS 2024 (USA)
- **Poster Presenter and Participant** in the MoML Conference & Hackathon, Valence Labs & Mila, 2024 (Canada)
- **Poster Presenter** at the OSU Research Showcase, Oregon State University, 2024 (USA)
- **Founder & Organizer** for Explainable AI reading group at Oregon State University, 2024 (USA)
- **Member & Speaker** at the Pervasive Personalized Intelligence Center (PPIC), 2023 (USA)
- **Participant** in the AgAID Hackathon, 2022 (USA)
- **Mentor** to three junior engineers, providing structured guidance in UI development and software testing, (USA)
- **Integration Testing Lead** across multiple Kingland projects, improving code coverage by *80%* and coordinating framework adoption across engineering teams in the US and China, 2020 (USA)
- **Honourable Mention** at the Kingland Poster Presentation, Iowa State University, 2018 (USA)
- **Chairperson** for student chapter of Computer Society of India (India)

Skills

Programming: Python, SQL, Java, C/C++, C#

Machine Learning: Statistical Modeling, Experiment Design, Model Evaluation, Explainable AI

Frameworks & Tools: PyTorch, TensorFlow, Keras, Pandas, NumPy, Scikit-learn

Systems: AWS, Docker, Terraform, Jenkins, Git, CI/CD Pipelines

Data: Large-scale data processing, feature engineering, analytics pipelines